

Note. Add references to AttendanceRegistrationServiceContractsLib, AttendanceRegisterServiceLib and ServiceModel in AttendanceRegisterServer. At the beginning of file Program.cs of AttendanceRegisterServer project, add the line "using System.ServiceModel;"

In this project, we also need to add App.Config file to specify the database-connection string as shown in Fig. 22.

Now, take the connection string specified in App.Config AttendanceRegistrationSystem project and put it in example code as given below (please do not copy and paste these lines):

```
<?xml version="1.0" encoding="utf-8"?>
<configuration>
<connectionStrings>
<add name="EmployeeAttendanceDBEntities"
```

```
connectionString="metadata=res://*/
AttendanceDataModel.csdl|res://*/
AttendanceDataModel.ssd|res://*/
AttendanceDataModel.msl;provider=System.
Data.SqlClient;provider
connection string="quote;
Data Source=SUNANDA-PC\SQLEXPRESS;
Initial Catalog=AttendanceDB;Integrated
Security=True;Pooling=False;
MultipleActiveResultSets=True"quote;
" providerName="System.Data.
EntityClient" />
</connectionStrings>
</configuration>
```

Set AttendanceRegisterServer as our start-up project by right-clicking on the project name and selecting that option and building the entire solution.

We are done with our server-side now, and we can communicate with the client-side. Run the server-side solution by pressing F5.

Client-server interaction

As mentioned earlier, we will write the logic for button clicks in Main-Window.xaml.cs in the client-side and we will also write the logic to get the RFID every time an RFID tag is read by the RFID reader. But before that, we need to keep our server-side up

and running. For doing so, we have to add a service reference to our client-side project of the service, which we have already created. Follow the steps given below to add Service Reference to the client-side project.

1. Right-click on References node in the client-side project, Attendance-ManagementSystemClient and click on Add Service Reference in it (refer Fig. 23). Then, in the address box, pass the address MetadataExchange endpoint (<http://localhost:4569/mexep>) that we have already mentioned in the server-side. Note that, in both client- and server-side, we use <http://localhost:4569/ep>, as this is the address of the endpoint where the server-side exposes its functionality through WCF and the client gets server-side information. Now, click Go. Make sure that the server-side application is running in the background. We can now see all services that are exposed through that endpoint and registered in the metadata.

Here, the name of the service reference is set as ServiceReference1. From now on, we can make use of this namespace to refer to our server-side.

2. Add two namespaces, System.IO.Ports and System.Threading, in the beginning of the application in Fig. 24.

3. Following is the code in Main-Window.xaml.cs for getting the RFID number each time the RFID tag is brought in the vicinity of the reader. The code is as listed below.

```
public MainWindow()
{
InitializeComponent();

/* Creation of a new thread which will be
* used to update the UI later */
th = new Thread
(new ThreadStart(MyThread));
th.Start();

//store all available port names
ports = SerialPort.GetPortNames();
StringBuilder portNames = new
StringBuilder();
int i = 1;

foreach (string portname in ports)
```

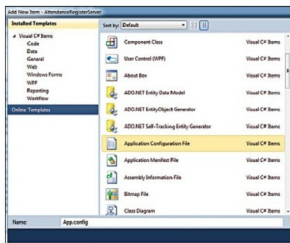


Fig. 22: Adding App.Config file from the application configuration file

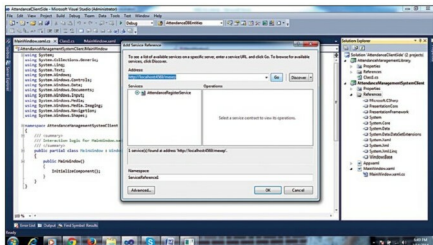


Fig. 23: Adding service reference